RCFX 5-Stage Counter-Current Fluidized Bed Heat Recovery System

Modeling-Only Evidence Package for Leverage (PERRY-RCFX-004 Rev 5.2)

Utility Patent Support Bundle | June 2026

Executive Summary

The RCFX is a 5-stage counter-current fluidized bed heat recovery architecture for low-pressure (0.1–0.5 bar) lunar/space ISRU operation. Iron shot (1.5–2.0 mm cold stages) serves dual roles as sensible heat media and collision agitator for cohesive regolith fines (Geldart C). All quantitative evidence is modeling-only (GPU DEM + empirical lumped model) with no hardware prototype. The package provides a narrow, honest, reproducible 'paper asset' suitable for grant proposals, partner discussions, or funding pitches to attract resources for the hardware validation phase.

Key Results (Clean Physical-Lid, Real Gas Drag Only)

Good-Variable Primary Evidence (physical_drag_real_u3.5_iron1.5mm_step002000.npz, N=6500, 100% inside physical lid+freeboard):

- 1.5 mm iron @ 3.5 m/s superficial (real drag, drag_mult=1.0, physical F-boundaries, e=0.95 clips)
- Iron mean height: 34.47 mm | Regolith mean height: 11.56 mm
- Effective Mobilization Index (EMI): 3.58× vs no-iron baseline (~3.23 mm bed, ~87% dead)
- Containment: 100.0% inside physical lid | Step: 2000 (real drag only)

Lumped Model Performance (five_stage_counterflow.py, vol_flow = U × AREA fixed, 100 kg/hr ref, 700 K ΔT):

- Overall thermal effectiveness: 75.6% at 0.14 bar reference
- Blower power: 221 W (1.88% parasitic on recovered heat)
- 3 explicit numbers: mdot_reg = 0.027778 kg/s; vol_flow = 0.0066 m^3/s per stage; A not defined in lumped (forward NTU in proper_ntu_thermal.py with Gunn h)

Differentiation: Static sintered distributor + passive weirs + iron collisions + physical lid. No continuous rotary dynamic seals or rotating mechanicals in the hot abrasive regolith path (unlike auger-based CoRHE concepts that require extensive prototype testing for seal/wear life).

Formal Drawings & Reproducibility

26 vector B&W drawings (SVG/PDF) in bundle: FIG.1 system, FIG.2 stage cross-section, FIG.3 iron agitation (updated with good-var data), FIG.4 counter-current, FIG.5 eff vs P, FIG.6 distributor, FIG.7 qualitative, S1–S4 sensitivities/scaling (for enablement). All traceable to clean evidence.

Reproducible kit: sims/custom_gpu_dem/highn_sensitivity.py (runner with --campaign/--n, auto base+add for >6500 particles, JSON reports, cell-list default); common/dem_kernels.py + optimized_step.py (physical mode, real rho_g, F-based boundaries); models/five_stage_counterflow.py + proper_ntu_thermal.py (3 numbers + honest forward NTU, no circular). Raw .npz: rung1_highn_checkpoints/ (incl. good-var), highn_sens_checkpoints/, rung0/rung5 500k locks. All 100% inside physical-lid data only.

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Honest Limitations (Full Candor – per 37 CFR 1.56 / MPEP 2001)

- No hardware prototype or physical test data exists. All numbers from modeling (lumped + GPU DEM).
- At exact reference point (0.14 bar / low U_G), bulk real gas drag alone is insufficient for reference iron sizes; the claimed static sintered distributor (sub-grid jet momentum in primary evidence) is essential (drag-fix verification + momentum budget: mdot_gas_single ≈ 0.000289 kg/s per stage too low for observed velocities).
- Viable real-drag points exist within envelope (e.g., 1.5 mm iron at 3.5 m/s as primary good-var).
- Any real system still requires standard boundary vessel seals + regolith feed/discharge interfaces (valves/locks) for vacuum/abrasive/thermal conditions. These are the same class of problems addressed in prior NASA ISRU work (PILOT, ROxygen, etc.) and require physical prototypes, simulant testing, and qualification.
- This testing is unaffordable with current resources. The project has no path to hardware development or commercialization without millions in additional funding for the prototype phase. The design was deliberately scoped to avoid continuous rotary dynamic seals in the abrasive path (the even harder problem in auger recuperators), but the boundary interface work remains.

Value as Leverage Asset

This is a complete, narrow, reproducible 'paper de-risking' package:

- Narrow claims (Option A) directly supported by specific good-var DEM (1.5 mm iron / 3.5 m/s real drag, exact metrics above).
- 75.6% effectiveness / <2% parasitic at low pressure with static hardware only (no mechanical stirrers).
- Formal B&W vector drawings + citable evidence summary (raw .npz traceable) suitable for inclusion in proposals or partner decks.
- Explicit MPEP 2164/2163/2001 + 37 CFR 1.56 language and candor (positive primary + verification limits disclosed).
- 'Run once and done' — no further simulation required for the current scope.

Ideal for: SBIR / NASA / ESA proposals (shows specific numbers, honest scope, reproducibility); discussions with hardware-capable partners or investors ('mechanism validated on paper at particle scale with clean physical-lid data; need collaborator for distributor/seal prototype validation and lunar simulant testing'). The bundle (spec + claims + drawings + index + evidence) + clean tar (~3.8 MB, no raw ckpts) is ready to hand off.

Bundle Location & Transfer

Core: rcfx/patent_application/2026-06-05/ (RCFX_Complete_Clean_Utility_Spec_and_Evidence.md, Suggested_Claims_for_Conversion.md, UTILITY_BUNDLE_INDEX.md, Patent_Citable_Evidence_Summary.md, Cover_Sheet_Info.txt, patent_drawings/ with 26 files incl. updated good-var FIG.3, PROJECT_STATUS.txt)
Evidence: rcfx/patent_evidence/2026-06-04/ (RCFX_Patent_Evidence_Package_2026-06-04.docx, COLD_CLAIMS_AND_MATH_REVIEW.md, exhibits A–E, audits, FILING_READINESS.md)
Clean tar (no checkpoints/pyc): /home/nick/rcfx-utility-bundle-2026-06-05.tar.gz (3.8 MB)
Reproducibility sources: models/ + sims/custom_gpu_dem/ (highn_sensitivity.py, kernels, physical mode steppers, clean .npz in rung1_highn_checkpoints/ and highn_sens_checkpoints/)

Git: c484fbc (utility cleanup + transfer prep) + later updates. All data 100% inside physical-lid, post-containment, raw .npz traceable. No new matter beyond PERRY-RCFX-004 Rev 5.2.

This report reviews the complete current state. The modeling package is finished and honest. It provides leverage as a scoped, traceable, professional asset for attracting the prototype funding required to advance. No further modeling changes the hardware barrier.